

Exercise 5.4

Create a function called `arithmetic_sum` that takes in integers `a`, `d`, and `n` and prints out the sum of the sequence, known as a series.

$$a + (a+d) + (a+2*d) + \dots + (a+(n-1)*d)$$

`a` is the first term, `d` is the difference between terms, and `n` is the number of terms in the series.

For example, the values `a = 1`, `d = 1` and `n = 100` is just the sum of the first 100 numbers.

$$1 + 2 + 3 + \dots + 99 + 100$$

For `a = 3`, `d = 2` and `n = 6` we would have,

$$3 + 5 + 7 + 9 + 11 + 13$$

One last example, `a = 10`, `d = 4` and `n = 5`

$$10 + 14 + 18 + 22 + 26$$

Calling the function with the following arguments (values) should result in the following.

function call	series	return value
<code>arithmetic_sum(1,1,100)</code>	<code>1 + 2 + ... + 99 + 100</code>	<code>5050</code>
<code>arithmetic_sum(3,2,15)</code>	<code>3 + 5 + ... + 29 + 31</code>	<code>255</code>
<code>arithmetic_sum(10,4,5)</code>	<code>10 + 14 + 18 + 22 + 26</code>	<code>90</code>
<code>arithmetic_sum(-5, -2, 20)</code>	<code>-5 -7 - ... - 41 - 43</code>	<code>-480</code>

You can copy the following into `main.py` to get started. NOTE: You will need to update the function signature.

```
def arithmetic_sum():
    pass

if __name__ == "__main__":
    print(arithmetic_sum(1,1,100)) # prints the return value - 5050
    print(arithmetic_sum(3,2,15)) # prints the return value - 255
    print(arithmetic_sum(10,4,5)) # prints the return value - 90
    print(arithmetic_sum(-5, -2, 20)) # prints the return value - -480
```